

# MH6822 - REGULATORY TECHNOLOGY

## Assignment 1: Designing a Jurisdiction-Aware RegTech Tool

*A fairness and explainability compliance prototype for credit scoring across the United States and the European Union*

| Field                      | Details to complete before submission                  |
|----------------------------|--------------------------------------------------------|
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| Selected regulated entity  | JPMorgan Chase & Co.                                   |
| Selected domain            | Credit scoring, fair lending, and algorithmic fairness |
| Selected jurisdictions     | United States and European Union                       |
| Selected format for Task 3 | Option A - Working Prototype (Python)                  |

### Executive Summary

This work designs a jurisdiction-aware RegTech tool named FairCredit Jurisdiction Engine. The tool is positioned for a hypothetical early-stage RegTech company serving the Chief Compliance Officer (CCO) and Model Risk teams of a global bank. The selected regulated entity is JPMorgan Chase & Co., a U.S.-based financial holding company with worldwide operations and meaningful non-U.S. activity in Europe. The selected regulatory domain is credit scoring and algorithmic fairness, because the United States and the European Union regulate AI-assisted credit decisions through materially different policy choices.

In the United States, the design is anchored in ECOA/Regulation B and CFPB guidance: the tool focuses on non-discrimination, statistically defensible credit scoring, and specific adverse-action explanations for consumers. In the European Union, the design is anchored in the AI Act: creditworthiness and credit-score AI systems are treated as high-risk uses, so the tool adds bias testing, technical documentation, human oversight, audit logging, and governance evidence. The same synthetic loan-application input therefore produces different compliance outputs depending on the active jurisdiction.

### Task 1 - Selection and Research

The assignment requires one real, named regulated entity with operations in the United States and at least one non-U.S. jurisdiction, and one specific regulatory domain such as fair lending, algorithmic fairness, credit scoring, model risk management, AML/KYC, reputational risk, or AI governance. This submission selects JPMorgan Chase & Co. and the domain of credit scoring / algorithmic fairness.

| Requirement           | Chosen answer        | Rationale                                     |
|-----------------------|----------------------|-----------------------------------------------|
| Real regulated entity | JPMorgan Chase & Co. | A major U.S.-based financial institution with |

|                       |                                         |                                                                                                                                                 |
|-----------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                                         | worldwide operations and regulated banking activities.                                                                                          |
| U.S. operations       | United States                           | JPMorgan Chase Bank, N.A. operates across the United States and is subject to U.S. banking, consumer-protection, and fair-lending requirements. |
| Non-U.S. jurisdiction | European Union                          | J.P. Morgan has significant European operations, and EU financial services activities create a credible cross-jurisdiction case.                |
| Regulatory domain     | Credit scoring and algorithmic fairness | The domain directly illustrates how algorithmic credit decisions can create divergent compliance duties across jurisdictions.                   |
| Tool format           | Option A - Working Prototype            | A lightweight Python prototype demonstrates jurisdiction configuration, synthetic inputs, different outputs, and a model card/governance stub.  |

### Entity rationale: JPMorgan Chase & Co.

JPMorgan Chase & Co. is a strong fit because it is a large, internationally active financial services firm. Its public reporting describes it as a financial holding company based in the United States, with global activity across consumer and community banking, commercial banking, corporate and investment banking, asset and wealth management, and other financial services. This makes it an appropriate example for a cross-border RegTech product because the bank must manage overlapping but different legal expectations in the United States and Europe. <sup>[1-2]</sup>

### Domain rationale: credit scoring and algorithmic fairness

Credit scoring is selected because it is not only a technical modelling problem but also a political and distributive problem. A credit model determines who receives access to credit, on what terms, and with what explanation. U.S. rules focus heavily on non-discrimination and specific adverse-action reasons under ECOA/Regulation B. EU rules treat creditworthiness and credit-score AI systems as high-risk AI when they determine access to financial resources or essential services. This creates an ideal use case for a jurisdiction-aware compliance tool because the tool must switch both its internal testing behaviour and its reporting outputs.

| Regulatory choice      | United States                                                                                                | European Union                                                                                                      | Design consequence for the tool                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Legal frame            | ECOA / Regulation B and CFPB adverse-action expectations                                                     | EU AI Act high-risk AI governance framework                                                                         | One tool cannot use the same checklist in both jurisdictions.                                |
| Core concern           | Specific reasons for denial and avoiding prohibited-basis discrimination                                     | Risk management, data governance, transparency, human oversight, lifecycle monitoring                               | US report emphasizes adverse-action reasons; EU report includes broader governance evidence. |
| Credit model treatment | Credit scoring systems must support fair and accurate decision processes, with adverse action notice duties. | AI systems evaluating creditworthiness or establishing credit scores are high-risk unless used for fraud detection. | EU mode triggers a high-risk AI compliance pack and audit trail.                             |
| Explanation            | Specific and accurate                                                                                        | Transparency and                                                                                                    | Both modes explain, but EU                                                                   |

|  |                                            |                                                    |                                                         |
|--|--------------------------------------------|----------------------------------------------------|---------------------------------------------------------|
|  | adverse-action explanation to the consumer | documentation obligations for high-risk AI systems | output also documents oversight and lifecycle controls. |
|--|--------------------------------------------|----------------------------------------------------|---------------------------------------------------------|

## Task 2 - Values Audit

Before designing the tool, this values audit sets out the values and commercial choices of the hypothetical RegTech provider. The answers are intentionally specific: the goal is not to perform virtue, but to reveal the real trade-offs behind a compliance product.

### 1. Mission Statement and Company Profile

Hypothetical company name: CivicFair RegTech Pte. Ltd.

**Mission statement:** CivicFair RegTech helps global financial institutions detect, explain, and govern algorithmic credit risk across jurisdictions, so that compliance teams can make defensible decisions without pretending that judgement has been outsourced to software.

**Core values:** (1) honesty about uncertainty, (2) explainability over cosmetic dashboards, (3) respect for affected consumers, and (4) practical usefulness for compliance officers under time pressure. The company is good at translating legal/regulatory differences into testable rules, audit logs, and model documentation. It is not a law firm and does not give legal opinions.

**Stage and scale:** CivicFair is a seed-stage company with 8 employees: two ML engineers, one data engineer, one product manager, one compliance analyst, one legal/regulatory researcher, one UX designer, and one founder/CEO. Its first-year target is to secure 3-5 pilot customers in banking or fintech, with annual recurring revenue of approximately USD 750,000 to USD 1.2 million. Its costs are concentrated in cloud infrastructure, regulatory research, model validation, and customer implementation. The initial geographic scope is the United States and the European Union; other jurisdictions would be added only after the rules engine and update workflow are mature.

### 2. Whose Perspective Does the Tool Serve?

The paying client is the Chief Compliance Officer and the model risk/compliance teams of a bank. The tool is therefore designed primarily for the CCO: it helps the CCO see whether a credit model is likely to create legal, reputational, or consumer-harm risk in the active jurisdiction. The regulator and consumer are secondary but important stakeholders. The regulator benefits because the tool generates auditable evidence. Consumers benefit if the tool catches biased outcomes or produces clearer reasons for credit denial.

There are real conflicts. The CCO may prefer a tool that reduces regulatory scrutiny and operational burden. A regulator may prefer more conservative testing and fuller disclosure. Consumers may prefer more accessible explanations and fewer false denials. These objectives can conflict: a stricter EU-style

fairness threshold may produce more false positives, increasing review costs for the bank, while a lighter U.S.-style workflow may be faster but may not provide enough evidence for an EU high-risk AI file.

The decision to serve the CCO first is commercial and competency-based. A CCO has budget and a concrete pain point: the cost of maintaining jurisdiction-specific model governance. CivicFair is also technically better at compliance workflow and risk reporting than at direct consumer-facing legal advice. However, the design deliberately keeps consumer harm visible by reporting false-negative risk and adverse-action explanation quality rather than only producing a green/red compliance badge.

### 3. Is the Tool Measuring Genuine Risk or Documenting Compliance?

The tool measures genuine risk and also documents compliance, but the design intentionally prioritises risk detection. A purely documentary tool would generate a report showing that each required control has an owner and a date. That is useful, but it can become a compliance theatre exercise.

Specific design choice: the EU mode includes a group-level approval-rate disparity test and a stability/drift indicator even where the immediate customer only asked for a documentation pack. This feature improves risk detection because it can reveal that a credit model has a materially lower approval rate for a protected or proxy-sensitive group, even if the model documentation is complete. The tool also refuses to show a final “approved for deployment” status when fairness metrics are missing or stale. This is intentionally inconvenient for the paying client, but it prevents the product from becoming merely a report generator.

### 4. Who Bears the Cost If the Tool Gets it Wrong?

| Failure                         | Specific harm                                                                                                      | Main stakeholder harmed                                                         | Mitigation in design                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| False positive                  | Tool flags a model as risky when the disparity is caused by sample noise or missing context.                       | Bank compliance/model teams; delayed product launch; unnecessary manual review. | Confidence notes, minimum sample size checks, and human override process.                                     |
| False negative                  | Tool misses a discriminatory pattern, so consumers are denied credit unfairly or receive weak explanations.        | Consumers first; regulator and bank later.                                      | Protected-group monitoring, proxy review, adverse-action explanation quality checks, and periodic re-testing. |
| Performance drift               | A model that was acceptable at validation begins performing differently as applicant population changes.           | Consumers and bank; drift creates both harm and regulatory exposure.            | Monthly drift dashboard and automatic stale-model warnings.                                                   |
| Jurisdictional misconfiguration | EU applications are checked under U.S. logic only, omitting high-risk AI documentation and oversight requirements. | Bank CCO and affected EU consumers.                                             | Mandatory jurisdiction selection, rule-version banner, and hard stop if jurisdiction metadata is missing.     |

|                         |                                                                                             |                                 |                                                                                                            |
|-------------------------|---------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------|
| Over-reliance by humans | Compliance staff treat software output as a legal conclusion and stop exercising judgement. | Bank, consumers, and regulator. | Output labels every result as decision support, not legal advice, and requires accountable human sign-off. |
|-------------------------|---------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------|

### Task 3 - Tool Design: FairCredit Jurisdiction Engine

Chosen format: Option A - Working Prototype. The prototype is a Python-based compliance demonstrator. It is not production-ready, but it demonstrates the core logic required by the assignment: jurisdiction configuration, simulated data input, outputs that change when the jurisdiction changes, and a model card/governance stub.

| Assignment requirement for Option A                        | How this submission satisfies it                                                                                                                                             |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jurisdiction configuration layer                           | A rules dictionary stores the active jurisdiction, regulatory focus, threshold policy, required governance fields, and output meaning.                                       |
| At least one live or simulated data input                  | Synthetic loan-application data is generated with credit score, income, debt-to-income ratio, delinquencies, requested loan amount, gender, and ethnicity.                   |
| Output means something different depending on active rules | US output focuses on adverse-action reasons and fair-lending monitoring; EU output adds high-risk AI documentation, human oversight, audit logging, and governance evidence. |
| Model card or governance stub                              | The script generates model_card.json describing purpose, assumptions, limitations, failure modes, intended use, and non-intended uses.                                       |

#### 3.1 Product Concept

FairCredit Jurisdiction Engine is a compliance-support tool for banks using machine-learning models in credit scoring or loan pre-approval. The tool takes a dataset, a trained or simulated credit model, and a jurisdiction code. It then runs the rule set relevant to that jurisdiction and produces a report for the compliance officer. The product does not replace legal advice or final human judgement. It is a structured decision-support system that helps the CCO understand which checks were performed, what risks were detected, and what remains outside the tool boundary.

#### 3.2 Regulatory Mechanics Encoded in the Tool

United States mode: The prototype treats U.S. compliance as centred on ECOA/Regulation B adverse-action explanation and non-discrimination. The CFPB has stated that when lenders use AI or complex models, consumers must receive accurate and specific reasons for credit denials, and there is no special exemption for artificial intelligence.<sup>[3]</sup>

European Union mode: The prototype treats EU compliance as a high-risk AI governance problem. The AI Act framework identifies creditworthiness evaluation and credit-score systems as high-risk because these decisions affect access to financial resources or essential services. High-risk AI systems require

controls such as risk management, data quality, documentation and traceability, transparency, human oversight, accuracy, cybersecurity, and robustness.<sup>[5-6]</sup>

| Tool module               | US mode behaviour                                                          | EU mode behaviour                                                                      |
|---------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Jurisdiction banner       | Shows Regulation B / ECOA and adverse-action focus.                        | Shows EU AI Act high-risk AI focus and governance-control requirements.                |
| Data intake               | Requires applicant-level score inputs and variables used for reason codes. | Requires score inputs plus audit metadata, governance controls, and monitoring period. |
| Fairness check            | Uses approval disparity as an internal risk indicator and review trigger.  | Uses approval disparity as part of high-risk AI risk management evidence.              |
| Explanation output        | Produces top adverse-action reasons for denied applicants.                 | Produces explanations plus documentation of oversight, logs, and human review status.  |
| Compliance decision label | REVIEW_REQUIRED where fairness/adverse-action review is needed.            | REVIEW_REQUIRED where fairness issue or high-risk AI control gap exists.               |
| Human review              | Required for proxy-risk warnings and adverse-action quality review.        | Required as a formal high-risk AI human oversight control.                             |

### 3.3 Prototype Architecture

Machine-learning component: The prototype trains a logistic regression classification model on synthetic loan-application data. The model predicts an approval probability for each applicant using income, credit score, debt-to-income ratio, delinquencies, and requested loan amount. Gender and ethnicity are deliberately excluded from training and retained only for post-model fairness auditing. I chose logistic regression because it is simple, interpretable, and appropriate for a classroom prototype. The tool therefore demonstrates both model-based credit scoring and jurisdiction-specific compliance monitoring.

| Layer                | Purpose                                                                                                         | Example component                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Input layer          | Receives synthetic credit-application dataset and selected jurisdiction.                                        | Generated pandas DataFrame + jurisdiction = "US" or "EU".                             |
| Rules engine         | Loads regulatory parameters and required checks.                                                                | JURISDICTION_RULES dictionary with jurisdiction-specific thresholds and requirements. |
| Model layer          | Scores applications using a simple, interpretable model.                                                        | Logistic regression classifier.                                                       |
| Risk analytics layer | Calculates approval disparity, demographic parity difference, false-negative rates, and adverse-action reasons. | compute_fairness_summary(), generate_adverse_reasons().                               |
| Report layer         | Produces jurisdiction-specific output.                                                                          | us_compliance_report.json or eu_compliance_report.json.                               |
| Governance layer     | Documents model assumptions, intended use, limitations, and failure modes.                                      | model_card.json.                                                                      |

### 3.4 Executable Python Prototype and How to Run It

The full executable prototype is submitted separately as Task3\_jurisdiction\_aware\_regtech\_tool.py. The Word document explains the design and actual output; the .py file contains the complete runnable implementation. Running the script generates three JSON files: us\_compliance\_report.json, eu\_compliance\_report.json, and model\_card.json.

#### Key implementation excerpt:

```
JURISDICTION_RULES = {
    "US": {
        "regulatory_focus": "Regulation B / ECOA adverse-action explainability and fair lending monitoring",
        "disparate_impact_min_ratio": 0.80,
        "requires_adverse_action_reasons": True,
        "requires_technical_documentation": False,
        "requires_human_oversight_record": False,
    },
    "EU": {
        "regulatory_focus": "EU AI Act high-risk AI system governance for creditworthiness assessment",
        "disparate_impact_min_ratio": 0.90,
        "requires_adverse_action_reasons": True,
        "requires_technical_documentation": True,
        "requires_human_oversight_record": True,
    },
}

MODEL_FEATURES = [
    "income", "credit_score", "debt_to_income", "delinquencies", "requested_loan_amount"
]

# Protected attributes are excluded from model training but retained for fairness audit.
AUDIT_ONLY_ATTRIBUTES = ["gender", "ethnicity"]
```

#### Run instructions:

```
pip install pandas numpy scikit-learn
python Task3_jurisdiction_aware_regtech_tool.py

# Optional validation of generated JSON files
python -m json.tool task3_outputs/us_compliance_report.json
python -m json.tool task3_outputs/eu_compliance_report.json
python -m json.tool task3_outputs/model_card.json
```

### 3.5 Actual Prototype Run Result

The prototype was successfully executed and generated three JSON output files: us\_compliance\_report.json, eu\_compliance\_report.json, and model\_card.json. The JSON files are valid and can be formatted using python -m json.tool. The submitted JSON files are the actual outputs from the prototype run.

| Output item            | Actual result                                                                | Interpretation                                                                                             |
|------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Model performance      | test_accuracy = 0.6861; test_auc = 0.6553; test_size = 360; train_size = 840 | The model is adequate for a classroom prototype, not for production credit decisioning.                    |
| Gender fairness metric | disparate impact ratio = 0.9107                                              | No fairness alert was triggered for gender because the ratio stayed above both internal review thresholds. |
| Ethnicity fairness     | disparate impact ratio = 0.6575                                              | This triggered review in both                                                                              |



|                            |                                                          |                                                                                                               |
|----------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| metric                     |                                                          | jurisdictions because it is below the U.S. threshold of 0.80 and the EU threshold of 0.90.                    |
| Sample denied applicant    | Applicant A00424; predicted approval probability = 0.089 | The model generated adverse-action reasons for a denied applicant.                                            |
| Top adverse-action reasons | delinquencies; debt_to_income; credit_score              | These reasons are used in the U.S.-style adverse-action explanation and also feed the EU documentation trail. |

The 0.80 U.S. and 0.90 EU thresholds are internal review triggers, not legal safe harbours. The U.S. threshold reflects a familiar disparate-impact screening convention used for internal fair-lending review. The EU threshold is deliberately more conservative because EU mode is not limited to adverse-action review; it is intended to support high-risk AI governance, lifecycle monitoring, and earlier escalation before deployment.

| Same run / same model        | US output meaning                                                                          | EU output meaning                                                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Prototype compliance status  | REVIEW_REQUIRED                                                                            | REVIEW_REQUIRED                                                                                                                        |
| Threshold applied            | Ethnicity disparate impact ratio 0.6575 is below the U.S. internal threshold of 0.80.      | Ethnicity disparate impact ratio 0.6575 is below the stricter EU internal threshold of 0.90.                                           |
| Main compliance focus        | Fair-lending monitoring and adverse-action explanation.                                    | High-risk AI governance, technical documentation, audit logging, and human oversight.                                                  |
| Jurisdiction-specific fields | adverse_action_notice_required = true;<br>consumer_facing_reason_codes generated.          | high_risk_ai_system = true;<br>technical_documentation_required = true; human_oversight_required = true;<br>audit_log_required = true. |
| Practical consequence        | Escalate to compliance team to check potential disparate outcomes and explanation quality. | Do not treat the model as ready for EU deployment until high-risk governance evidence and fairness mitigation are reviewed.            |

This result is useful because it proves that the tool's jurisdiction logic is substantive rather than decorative. The same data and model produce different compliance packages: the U.S. report is centred on fair-lending monitoring and adverse-action reasons, while the EU report includes high-risk AI governance fields. The legal conclusion is not automated; the tool only identifies review triggers and missing governance evidence for accountable human review.

#### Actual generated file inventory:

- task3\_outputs/us\_compliance\_report.json - U.S. Regulation B / ECOA style report.
- task3\_outputs/eu\_compliance\_report.json - EU AI Act high-risk AI style report.
- task3\_outputs/model\_card.json - governance documentation stub / model card.



### 3.6 Model Card / Governance Documentation Stub

| Field                | Actual model card content / interpretation                                                                                                                                                  |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model name           | Synthetic Credit Approval Logistic Regression Model                                                                                                                                         |
| Model type           | Logistic regression classification model.                                                                                                                                                   |
| Purpose              | Classroom prototype for jurisdiction-aware RegTech compliance monitoring in credit scoring / algorithmic fairness.                                                                          |
| Not intended for     | Real credit decisions, production lending, legal advice, or full replacement of human compliance judgement.                                                                                 |
| Input variables      | income, credit_score, debt_to_income, delinquencies, requested_loan_amount.                                                                                                                 |
| Protected attributes | gender and ethnicity are excluded from model training and used only for fairness auditing.                                                                                                  |
| Performance          | test_accuracy = 0.6861; test_auc = 0.6553.                                                                                                                                                  |
| Assumptions          | Synthetic data is used because real bank credit data is unavailable; protected attributes are retained only for audit; jurisdictions require different reporting and governance outputs.    |
| Known limitations    | Synthetic data may not match a live applicant population; coefficient explanation is approximate, not causal; the tool covers fairness monitoring rather than the full EU AI Act lifecycle. |

### 3.7 Failure Mode Analysis

| Scenario                        | What can go wrong                                                              | Tool response                                                              | Residual limitation                                      |
|---------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------|
| Rule changes mid-period         | The active legal requirement changes after a report was generated.             | Version every rule set and show rule-version date on report.               | The prototype does not automatically scrape law updates. |
| Model drift                     | The applicant population changes and fairness/performance metrics deteriorate. | Schedule periodic drift monitoring and retraining review.                  | Synthetic data cannot prove real-world drift behaviour.  |
| Jurisdictional misconfiguration | EU portfolio is checked using U.S.-only report settings.                       | Use validated jurisdiction codes and require compliance-owner approval.    | Human and process controls remain necessary.             |
| Proxy discrimination            | Protected attributes are excluded, but other variables act as proxies.         | Monitor group-level outcomes and flag significant disparities.             | Statistical flags do not prove legal discrimination.     |
| Explanation overclaim           | Reason codes are treated as causal explanations.                               | Label explanations as model-based approximations and require human review. | The prototype does not provide causal or legal proof.    |
| Human ignores warning           | A compliance officer deploys anyway.                                           | Audit log records warning and accountable sign-off.                        | Governance culture cannot be automated.                  |

### Task 3 One-Page Plain-Language Summary

FairCredit Jurisdiction Engine is a prototype RegTech tool for banks that use AI or statistical models in credit scoring. The tool solves a practical problem: a global bank cannot treat a credit scoring model as if the same compliance checklist applies everywhere. In the United States, the tool checks whether the decision process creates fair-lending concerns and whether denied consumers receive specific, accurate adverse-action reasons. In the European Union, the same credit scoring use case is treated as a high-risk AI governance problem, so the tool adds broader testing and documentation: bias metrics, technical documentation, human oversight, audit logging, and monitoring.

In the actual prototype run, both the U.S. and EU reports returned REVIEW\_REQUIRED because the ethnicity-based disparate impact ratio was 0.6575. However, the meaning of the alert differed by jurisdiction. The U.S. report treated it as a fair-lending and adverse-action review issue. The EU report treated it as part of a high-risk AI governance review requiring technical documentation, audit logging, and human oversight. This demonstrates that the tool does not merely change labels; it changes thresholds, report fields, and the governance evidence expected from the compliance team.

The product is right for CivicFair RegTech because the company is small and specialised. It does not try to replace enterprise governance platforms or law firms. Instead, it turns a narrow but important regulatory divergence into a repeatable workflow. This is more useful than a memo or spreadsheet because the tool changes its checks and output automatically when the jurisdiction changes. A memo can describe the difference once; a rules engine can apply the difference to each model run, each dataset version, and each deployment decision.

The main design boundary is deliberate: the tool does not decide whether a consumer should receive credit, and it does not provide a legal opinion. It supports a CCO by highlighting risks and missing controls. Human judgement remains necessary, especially when a rule changes, a model drifts, or two jurisdictions pull the bank in different directions. If more time and data were available, the next version would use real loan-performance data, more rigorous bias tests, SHAP/LIME explanations, an automated regulation-update workflow, and a better human-review interface.

## References

- [1] JPMorgan Chase & Co. Annual Report 2025. JPMorgan Chase & Co.  
<https://www.jpmorganchase.com/content/dam/jpmc/jpmorgan-chase-and-co/investor-relations/documents/annualreport-2025.pdf>
- [2] JPMorgan Chase Annual Report & Proxy page. JPMorgan Chase & Co.  
<https://www.jpmorganchase.com/ir/annual-report>
- [3] CFPB Circular 2022-03: Adverse action notification requirements in connection with credit decisions based on complex algorithms. Consumer Financial Protection Bureau.  
<https://www.consumerfinance.gov/compliance/circulars/circular-2022-03-adverse-action-notification-requirements-in-connection-with-credit-decisions-based-on-complex-algorithms/>
- [4] 12 CFR § 1002.2 - Regulation B definitions. Consumer Financial Protection Bureau.  
<https://www.consumerfinance.gov/rules-policy/regulations/1002/2>
- [5] Recital 58 - EU AI Act Service Desk. European Commission AI Act Service Desk. <https://ai-act-service-desk.ec.europa.eu/en/ai-act/recital-58>
- [6] Navigating the AI Act - high-risk AI systems obligations. European Commission, Shaping Europe's Digital Future. <https://digital-strategy.ec.europa.eu/en/faqs/navigating-ai-act>
- [7] Regulation (EU) 2024/1689 official text. EUR-Lex. <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>
- [8] Assignment 1 instruction document uploaded by student. MH6822 Regulatory Technology. Information on Assignment 1.pdf
- [9] Actual Task 3 prototype run outputs generated by Task3\_jurisdiction\_aware\_regtech\_tool.py: us\_compliance\_report.json, eu\_compliance\_report.json, and model\_card.json.